37. 'Dry Ice' is –

(a) Frozen ice (b) Frozen Carbon dioxide
(c) Frozen water (d) Frozen oxygen

Uttarakhand U.D.A./L.D.A. (Mains) 2006
R.A.S./R.T.S. (Pre) 2012

Ans. (b)

See the explanation of above question.

38. 'Dry Ice' is –

(a) Vapour
(b) Ice at 0°C
(c) Solid CO_2
(d) By the reaction of Calcium Chloride, free water is tread as ice.

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (c)

See the explanation of above question.

39. Water gas is –

(a) $CO + H_2$ (b) $CO + H_2O$
(c) $CO_2 + H_2$ (d) $CO_2 + H_2O$
(e) None of these

Chhattisgarh P.C.S. (Pre) 2016

Ans. (a)

Water gas is a synthesis gas with the mixture of carbon monoxide (CO) and hydrogen (H_2). Methyl alcohol is formed by water gas ($CO + H_2$) –

$$CO + 2H_2 \xrightarrow{\text{Catalyst}} CH_3OH$$

40. The gas, which comes out on opening a soda water bottle, is :

(a) Carbon dioxide (b) Hydrogen
(c) Nitrogen (d) Sulfur dioxide
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (a)

Soda water or carbonated water is water that has been infused with carbon dioxide gas under pressure. The gas, which comes out on opening a soda water bottle, is carbon dioxide that produces a bubbly drink.

41. What is the bond order of CO group?

(a) 1 (b) 2.5
(c) 3.5 (d) 3
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

In Molecular orbital theory the bond order formula can be defined as half of the difference between the number of electrons in bonding orbitals and antibonding orbitals.

Bond order = $1/2\ [N_b - N_a]$

Where,

N_b = number of bonding electrons

N_a = number of antibonding electrons

CO (Carbon monoxide) molecule has 10 valence electrons, four from carbon atom ($2s^2 2p^2$) and six from oxygen atom ($2s^2 2p^4$). According to molecular orbital diagram, molecular orbital configuration is given as

$(\sigma 2s)^2 (\sigma^* 2s)^2 (\sigma 2p_z)^2 (\pi 2p_x)^2 (\pi 2p_y)^2$

here, $N_b = 8$; $N_a = 2$

Thus, Bond order of CO = $1/2[8 - 2] = 3$

Note : Here the calculations are done ignoring the 1s orbitals because there will be 2 electrons in the bonding and 2 in the antibonding.

B. Hydrogen and its Compounds

Notes

- Hydrogen is a chemical element with symbol H, atomic number 1 and its electronic configuration is 1s.
- It is the lightest element in the periodic table with a standard atomic weight of 1.008.
- It occupies a unique position in the periodic table.
- In its properties, it behaves like alkali metals (Li, Na, K etc.) as well as halogens (F, Cl, Br etc.).

Isotopes of Hydrogen :

- There are three isotopes of hydrogen –

(i) **Protium -** ${}_1H^1$ (A = 1) Without any neutron – stable form
(ii) **Deuterium -** ${}_1H^2$ (A = 2) With one neutron – stable form
(iii) **Tritium -** ${}_1H^3$ (A = 3) With two neutrons – it is unstable and therefore it is radioactive. Like all radioactive isotopes tritium decays. As it decays it gives off or emits beta radiation.

- In nature, tritium is found in very less amount.
- It is prepared in atomic reactors by artificial techniques.
- Deuterium was discovered (1931) by American chemist-Harold C. Urey (for which he was awarded the Nobel Prize for Chemistry in 1934) and his associates F. Brickwedde and G. Murphy.
- The existence of tritium was established by Bleakney and Gould in 1934. Tritium was first produced in 1934 by

Ernest Rutherford, M.L. Oliphant and Paul Harteck - when they bombarded deuterium with high energy deuterons (nuclei of deuterium atoms).

Properties of Deuterium :

- It is an isotope of hydrogen.
- It is colourless, odourless and tasteless gas.
- It is insoluble in water.
- Its molecule is diatomic.
- Deuterium is now prepared largely by the electrolysis of heavy water.

Heavy water (D_2O) :

- It is the oxide of deuterium.
- It was discovered by Harold C. Urey in 1931 and G.N. Lewis was able to isolate the first sample of heavy water in 1933.
- It is a valuable substance. Its uses are as follows–

(i) **As neutron moderator** – The substance which moderates the fast moving neutrons in atomic reactors is termed as moderator. Heavy water is used as a neutron moderator in atomic reactors. It is also used as a coolant in reactors.

(ii) It is used to prepare deuterium and deuterium compounds.

Effect of Heavy Water on livings –

(i) Concentrated heavy water is harmful to body. It reacts slowly than ordinary water, as a result the physiological reactions of the body become abnormal.

(ii) It inhibits the growth of plants.

(iii) Seed germination stops in the presence of heavy water.

Manufacturing of Heavy water in India –

- The first heavy water plant was set up in India at **Nangal** (Punjab) in 1962. Other heavy water plants are at **Baroda** (Gujrat), **Tuticorin** (Tamil Nadu). **Kota** (Rajasthan), **Hazira** (Gujarat), **Manuguru** (Telangana).

Soft and Hard Water :

Soft water -

- It is treated water in which the only cation is sodium. It is free from dissolved salts of calcium or magnesium.
- It gives froth easily with soap.

Hard water :

- The water that contains an appreciable quantity of dissolved minerals of calcium and magnesium (calcium carbonate, magnesium carbonate, calcium chloride, magnesium chloride, etc.).
- The hardness of water depends on the solubility of different minerals of calcium and magnesium.
- Soap is the soluble sodium salt of **Stearic acid** ($C_{17}H_{35}COOH$).
- Hard water does not give froth easily with soap.
- **Rainwater** is naturally soft. When it goes underneath the ground, it picks up minerals like chalk, lime and mostly of calcium and magnesium salts.

Hydrogen Peroxide :

- It is a chemical compound with chemical formula H_2O_2.
- It was discovered by Louis Jacque Thenard in 1818.

Properties of Hydrogen Peroxide –

- It is pale blue - green liquid which appears colourless in a dilute solution, slightly more viscous than water.
- It is soluble in water, alcohol and ether. It is a weak acid.
- Its relative density is 1.47.
- Its boiling point is 150.2°C and freezing point is – 0.43°C.

Uses of Hydrogen Peroxide –

- Hydrogen peroxide is a natural antiseptic and germicide, therefore one of its most common uses is to clean wounds to prevent infection.
- It is used for bleaching silk, wool, hair and ivory.
- It is used in the preservation of milk, wine, etc.
- It is used as fuel or used as an oxidizer, with other fuels.

Question Bank

1. An element X has four electrons in its outermost orbit. What will be the formula of its compounds with Hydrogen?

(a) X_4H (b) X_4H_4

(c) XH_3 (d) XH_4

43rd B.P.S.C. (Pre) 1999

Ans. (d)

The compounds of element X which has four electrons in its outermost orbit will be XH_4.

$$\begin{array}{c} H \\ | \\ H-X-H \rightarrow XH_4 \\ | \\ H \end{array}$$

2. According to weight, the percentage of Hydrogen in water (H_2O), is –

(a) 44.45 % (b) 5.55 %

(c) 88.89 % (d) 11.11 %

Uttrakhand U.D.A./L.D.A. (Mains) 2006

Ans. (d)

Percentage of composition is the percent by mass of each element present in a compound.

In Water, H_2O = 2+ 16 = 18 g/mol

No. of Hydrogen atoms present in Water = 2

$\therefore$ Percentage of Hydrogen in water $= \frac{2}{18} \times 100 = 11.11\%$

3. One mole of hydrogen gas burns in excess of oxygen to give 290 KJ of heat. What is the amount of heat produced when 4g of hydrogen gas is burnt under the same conditions?

(a) 145 Kj (b) 290 Kj
(c) 580 Kj (d) 1160 Kj

I.A.S. (Pre) 1994

Ans. (c)

1 Mole of H_2 (Hydrogen) = 2 gm

$\therefore$ Energy released by burning 4 gm of hydrogen = 290×2
= 580 Kj

4. Burning of hydrogen produces –

(a) Oxygen (b) Ash
(c) Soil (d) Water

47th B.P.S.C. (Pre) 2005

Ans. (d)

Hydrogen gas is highly flammable and burns in air at a very wide range of concentrations between 4% to 75% by volume. Hydrogen gas cannot burn in absence of air . But by burning with oxygen it produces water.

5. Which one of the following fuels causes minimum air pollution?

(a) Kerosene oil (b) Hydrogen
(c) Coal (d) Diesel

U.P. Lower Sub. (Pre) 2015

Ans. (b)

Hydrogen is the purest combustion fuel. Water is generated from burning of hydrogen. While coal, kerosene oil and diesel are known as fossil fuel or carbonic fuel which generates carbon dioxide and many other harmful gases when it burns.

6. How is Hydrogen stored physically?

(a) In form of hydrides
(b) In form of water
(c) By compressing hydrogen gas
(d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (c)

Hydrogen can be stored physically as either a compressed gas or a liquid. Compressed hydrogen storage (which is most common) typically involves compressing hydrogen gas into high-pressure cylinders or tanks. The pressure is typically between 350 and 700 bar (around 5,000 and 10,000 psi). This compression increases the density of the hydrogen, allowing it to be stored in a smaller volume. Hydrogen can also be stored in a liquid state at cryogenic temperatures, which are significantly below the boiling point of hydrogen. This requires specialized insulated tanks to maintain the low temperatures. Storage of hydrogen as a liquid requires cryogenic temperatures because the boiling point of hydrogen at one atmosphere pressure is around – 252.9°C. Hydrogen can also be stored on the surfaces of solids (by adsorption) or within solids (by absorption). Cryo-Compressed Hydrogen is a hybrid method that stores hydrogen at cryogenic temperatures while also pressurizing it in a special vessel. This can achieve higher densities than conventional compressed or liquid hydrogen storage.

7. Heavy water is a type of –

(a) Coolant (b) Moderator
(c) Ore (d) Fuel

U.P.P.C.S. (Pre) 1993

Ans. (a) & (b)

Heavy water (D_2O) or deuterium oxide is used as a moderator as well as a coolant in nuclear reactors because it slows down neutrons effectively and also has a low probability of absorption of neutrons. Deuterium is an isotope of hydrogen which comprises both a neutron and a proton. Deuterium or heavy hydrogen reacts with oxygen to form deuterium oxide (D_2O), also known as heavy water. Normal water (H_2O) also used as a moderator as well as a coolant in nuclear reactors.

8. The chemical formula of Heavy Water is :

(a) H_2O (b) D_2O
(c) H_2CO_3 (d) H_2S

Jharkhand P.C.S. (Pre) 2003

Ans. (b)

See the explanation of above question.

9. The chemical formula for heavy water is :

(a) H_2O (b) N_2O
(c) D_2O (d) CuO
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

See the explanation of above question.

10. The substance used as moderator and coolant both, in nuclear reactors is :

(a) Ordinary water (b) Heavy water
(c) Liquid Ammonia (d) Liquid Hydrogen

U.P. P.C.S. (Pre) 2016

Ans. (a) & (b)

See the explanation of above question.

11. What is heavy water ?

(a) Oxygen + Heavy Hydrogen
(b) Hydrogen + Oxygen
(c) Hydrogen + New Oxygen
(d) Heavy Hydrogen + Heavy Oxygen

M.P.P.C.S. (Pre) 1991

Ans. (a)

See the explanation of above question.

12. Heavy water is that water –

(a) The temperature of which is kept constant at 4°C
(b) In which insoluble salts of Calcium and Potassium are present
(c) In which isotopes takes place of Hydrogen
(d) In which isotopes takes place of Oxygen

41st B.P.S.C. (Pre) 1996

Ans. (c)

See the explanation of above question.

13. Heavy water –

(a) Contains more dissolved air
(b) Contains deuterium in place of Hydrogen
(c) Contains more dissolved minerals and salts
(d) Contains organic impurities

U.P.P.C.S. (Mains) 2007

Ans. (b)

See the explanation of above question.

14. Who among the following discovered heavy water?

(a) Heinrich Hertz (b) H.C. Urey
(c) G.Mendel (d) Joseph Priestly

I.A.S. (Pre) 2008

Ans. (b)

Harold Clayton Urey, an American physical chemist, discovered deuterium and heavy water. He was awarded Nobel Prize in Chemistry in 1934 for his discovery of heavy hydrogen (deuterium). Heavy water is the oxide of deuterium, an isotope of hydrogen with one proton and one neutron in its nucleus.

15. Heavy water has molecular weight :

(a) 18 (b) 20
(c) 36 (d) 54

U.P. Lower Sub. (Mains) 2015

Ans. (b)

Heavy water (D_2O), also called deuterium oxide, is water composed of deuterium, a hydrogen isotope with mass double that of ordinary hydrogen. Thus, heavy water has a molecular weight of about 20, whereas the ordinary water has a molecular weight of about 18.

16. Consider the following statements :

Hard water is not suitable for

1. Drinking
2. Washing clothes with soap
3. Use in boilers
4. Irrigating crops

Which of these statements are correct ?

(a) 1 and 2 (b) 2 and 3
(c) 1, 2 and 4 (d) 1, 2, 3 and 4

I.A.S. (Pre) 2000

Ans. (d)

Hard water is described as 'hard' due to the presence of highly dissolved minerals specifically sulphates and chlorides of calcium and magnesium. Hard water is salty and therefore not suitable for drinking. It is very difficult to wash clothes with hard water as it requires more soap and leaves a messy scum that cannot be washed out easily. When hard water is boiled at home or in industries, it leaves deposits of calcium and magnesium salts. These deposits reduce the efficiency of boilers, kettles and pipes and can cause blockages and even burst of the boilers. If salt level increases in irrigation water, it becomes harder for lawns and landscape plant to take up water even though the soil is moist. Hard water blocks the xylem tissues of the plants and thus not suitable for irrigation.

17. Permanent hardness of water is due to –

(a) Chlorides and sulphates of Calcium and Magnesium
(b) Calcium bicarbonate sulphates
(c) Magnesium bicarbonate
(d) Chlorides of Silver and Potassium

Uttarakhand P.C.S. (Pre) 2005
40th B.P.S.C. (Pre) 1995

Ans. (a)

See the explanation of above question.

18. **The pH-value for water is –**
(a) Nearly zero (b) Nearly 7
(c) 5 or less than 5 (d) 8.7 or more
Uttarakhand P.C.S. (Pre) 2005

Ans. (b)

The pH value of pure water is 7. Pure water is neutral by nature. The solution with a pH less than 7 are said to be acidic and solutions with a pH greater than 7 are basic or alkaline.

19. **The pH value of water is**
(a) 4 (b) 7
(c) 12 (d) 18
(e) None of the above/More than one of the above
66th B.P.S.C. (Pre) (Re. Exam) 2020
65th B.P.S.C. (Pre) 2019

Ans. (b)

See the explanation of above question.

20. **Which is the purest form of water?**
(a) Tap water (b) Sea water
(c) Rainwater (d) Distilled water
M.P.P.C.S. (Pre) 2000

Ans. (c)

Among the given options rainwater is the purest form of water. The water on the earth is mixed with alkaline and acidic materials that make the water impure.

21. **Water is a good solvent of ionic salts because –**
(a) It has a high boiling point
(b) It has a high dipole moment
(c) It has a high specific heat
(d) It has no colour
I.A.S. (Pre) 1994

Ans. (b)

Water is a good solvent due to its polarity (it is dipolar in nature) and high dipole moment which can easily dissolve polar compounds like ionic salts. Water dissolves ionic salts by hydrating their component ions. For example, water dissolves NaCl by hydrating and stabilizing the Na^+ and Cl^- ions.

22. **Water can dissolve more substances than any other liquid because :**
(a) it is dipolar in nature
(b) it is a good conductor of heat
(c) it has high value of specific heat
(d) it is an oxide of hydrogen
I.A.S. (Pre) 2021

Ans. (a)

See the explanation of above question.

23. **Large quantities of drinking water is prepared from impure water by –**
(a) Desalination (b) Distillation
(c) Ion-exchange (d) Decantation
Uttarakhand P.C.S. (Pre) 2005

Ans. (a)

Desalination is a process that removes mineral components from saline water (also refer to removal of salts and minerals). Saltwater is desalinated to produce water suitable for human consumption or irrigation. Seawater desalination is a very effective way of production of potable water for drinking, irrigation and industries.

24. **Which gas is used in the purification of drinking water?**
(a) Helium (b) Chlorine
(c) Fluorine (d) Carbon dioxide
Jharkhand P.C.S. (Pre) 2013

Ans. (b)

Chlorine is presently an important chemical for water purification (such as in water treatment plants), as disinfectants and in bleach. Chlorine is usually used to kill bacteria and other microbes in drinking water supplies and public swimming pools.

25. **By which process the sea water can be converted into pure water?**
(a) Deliquescence (b) Efflorescence
(c) Electric separation (d) Reverse osmosis
R.A.S./R.T.S.(Pre) 2008

Ans. (d)

Reverse osmosis (RO) is a water purification technology that uses a semipermeable membrane to remove larger particles from drinking water. It is used in desalination of sea water or salty water. The most widely used method of desalination on a large scale involves reverse osmosis. If pure water and salty water are placed on both sides of a semipermeable membrane, water will flow towards the salty water side. If a high pressure (higher than the osmotic pressure) is applied to the salty water side of the semi-permeable membrane, the water will flow in the opposite direction and pure water will be obtained. Reverse osmosis can remove many types of molecules and ions from solutions including bacteria and is used in both industrial processes and the production of potable pure water.

26. **The process used for transforming salty water into pure water is called –**
(a) Deliquescence (b) Efflorescence